Ex. No: 15 **Mini Project**

DATE:04/10/2026 **Student Academic Performance Analysis**

Aim

To develop a Student Performance Analysis System using SQL and SQLite to store, manage, analyze, and evaluate student academic performance based on marks, study time, attendance, and performance trends.

Software Used

- Github Codespaces
- SQLite
- Oracle Database
- SQL
- PL/SQL

Description

The Student Performance Analysis System is developed to manage and analyze student academic records. The system stores student details, school information, study time, absences, and examination grades.

SQL queries are used to retrieve and analyze student performance data, while PL/SQL functions and procedures are used to classify students, generate reports, and evaluate academic progress. The system helps identify high-performing students and students who require academic support.

Database Tables

1. STUDENT_PERFORMANCE

Column Name	Data Type	Constraint
STUDENT_ID	NUMBER	PRIMARY KEY
STUDENT_NAME	VARCHAR2(30)	NOT NULL
SCHOOL	VARCHAR2(30)	NOT NULL
AGE	NUMBER	NOT NULL
STUDY_TIME	NUMBER	NOT NULL
ABSENCES	NUMBER	NOT NULL
G1	NUMBER	NOT NULL
G2	NUMBER	NOT NULL

G3	NUMBER	NOT NULL

Dataset: UCI Student Performance

Table: STUDENT_PERFORMANCE

Problem statement: Develop a student performance analysis system to evaluate academic marks, studytime, absences, and performance trends.

SQL queries:

1.Display student records with school, age, study time, absences, and grades.

2.Calculate the average final grade for each school.

3.Identify students who scored at least 15 in the final examination.

4.Display students with more than 10 absences.

5.Calculate the difference between first-period and final grades for each student.

PL/SQL tasks:

1.Create a function to classify students based on final examination marks.

2.Create a procedure to display the academic performance of a student using student ID.

3.Create a procedure to generate a school-wise performance report and identify students requiring academic support.

TABLE: STUDENT_PERFORMANCE

INPUT: create table student_performance(
student_id number primary key,
student_name varchar2(50),
school varchar2(50),
age number,
study_time number,
absences number,
g1 number,
g2 number,
g3 number

);

OUTPUT:

Table STUDENT_PERFORMANCE created.

INSERT VALUES 25

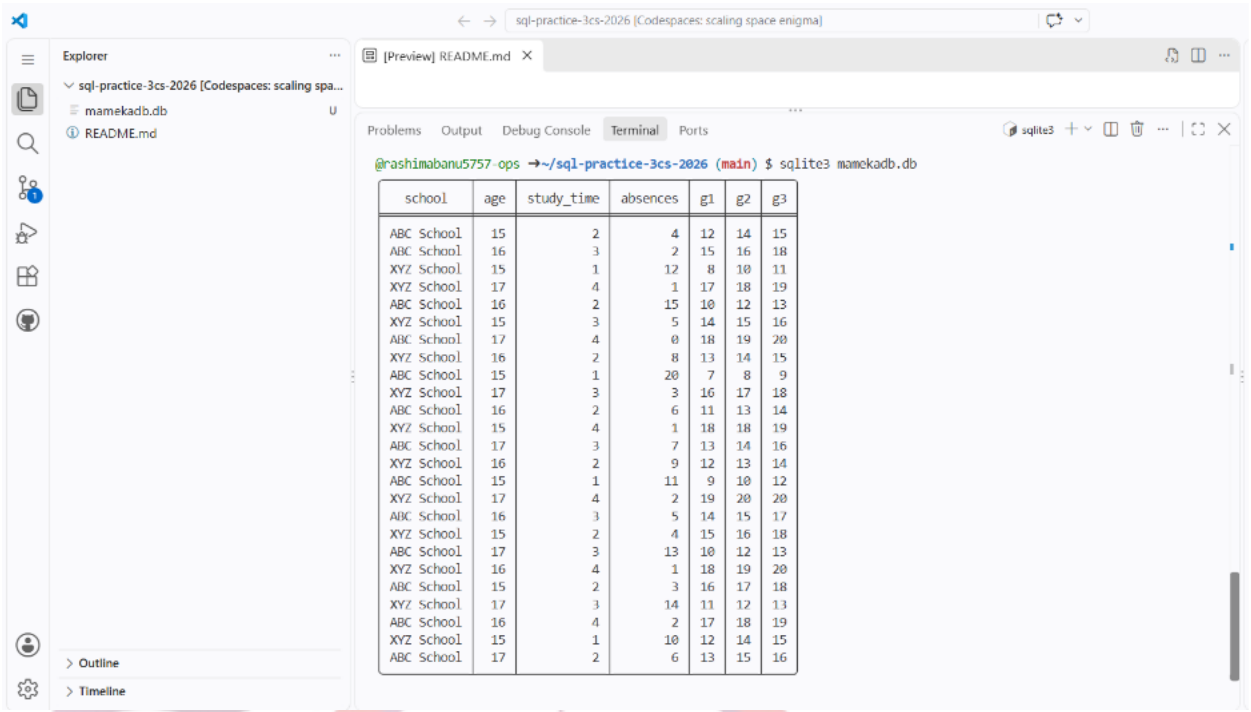
```
insert into student_performance values(101,'Arun','ABC School',15,2,4,12,14,15);
insert into student_performance values(102,'Priya','ABC School',16,3,2,15,16,18);
insert into student_performance values(103,'Rahul','XYZ School',15,1,12,8,10,11);
insert into student_performance values(104,'Divya','XYZ School',17,4,1,17,18,19);
insert into student_performance values(105,'Kiran','ABC School',16,2,15,10,12,13);
insert into student_performance values(106,'Meena','XYZ School',15,3,5,14,15,16);
insert into student_performance values(107,'Vijay','ABC School',17,4,0,18,19,20);
insert into student_performance values(108,'Anu','XYZ School',16,2,8,13,14,15);
insert into student_performance values(109,'Ravi','ABC School',15,1,20,7,8,9);
insert into student_performance values(110,'Sita','XYZ School',17,3,3,16,17,18);
insert into student_performance values(111,'Kumar','ABC School',16,2,6,11,13,14);
insert into student_performance values(112,'Deepa','XYZ School',15,4,1,18,18,19);
insert into student_performance values(113,'Hari','ABC School',17,3,7,13,14,16);
insert into student_performance values(114,'Latha','XYZ School',16,2,9,12,13,14);
insert into student_performance values(115,'Manoj','ABC School',15,1,11,9,10,12);
insert into student_performance values(116,'Nisha','XYZ School',17,4,2,19,20,20);
insert into student_performance values(117,'Prakash','ABC School',16,3,5,14,15,17);
insert into student_performance values(118,'Rekha','XYZ School',15,2,4,15,16,18);
insert into student_performance values(119,'Saravanan','ABC School',17,3,13,10,12,13);
insert into student_performance values(120,'Uma','XYZ School',16,4,1,18,19,20);
insert into student_performance values(121,'Vani','ABC School',15,2,3,16,17,18);
insert into student_performance values(122,'Yogesh','XYZ School',17,3,14,11,12,13);
insert into student_performance values(123,'Ajith','ABC School',16,4,2,17,18,19);
insert into student_performance values(124,'Bhavya','XYZ School',15,1,10,12,14,15);
insert into student_performance values(125,'Charan','ABC School',17,2,6,13,15,16);
```

SQL QUERIES

QUESTION: 1. Display student records with school, age, study time, absences, and grades :

SQL> select school,age,study_time,absences,g1,g2,g3 from student_performance;

OUTPUT:



school	age	study_time	absences	g1	g2	g3
ABC School	15	2	4	12	14	15
ABC School	16	3	2	15	16	18
XYZ School	15	1	12	8	10	11
XYZ School	17	4	1	17	18	19
ABC School	16	2	15	10	12	13
XYZ School	15	3	5	14	15	16
ABC School	17	4	0	18	19	20
XYZ School	16	2	8	13	14	15
ABC School	15	1	20	7	8	9
XYZ School	17	3	3	16	17	18
ABC School	16	2	6	11	13	14
XYZ School	15	4	1	18	18	19
ABC School	17	3	7	13	14	16
XYZ School	16	2	9	12	13	14
ABC School	15	1	11	9	10	12
XYZ School	17	4	2	19	20	20
ABC School	16	3	5	14	15	17
XYZ School	15	2	4	15	16	18
ABC School	17	3	13	10	12	13
XYZ School	16	4	1	18	19	20
ABC School	15	2	3	16	17	18
XYZ School	17	3	14	11	12	13
ABC School	16	4	2	17	18	19
XYZ School	15	1	10	12	14	15
ABC School	17	2	6	13	15	16

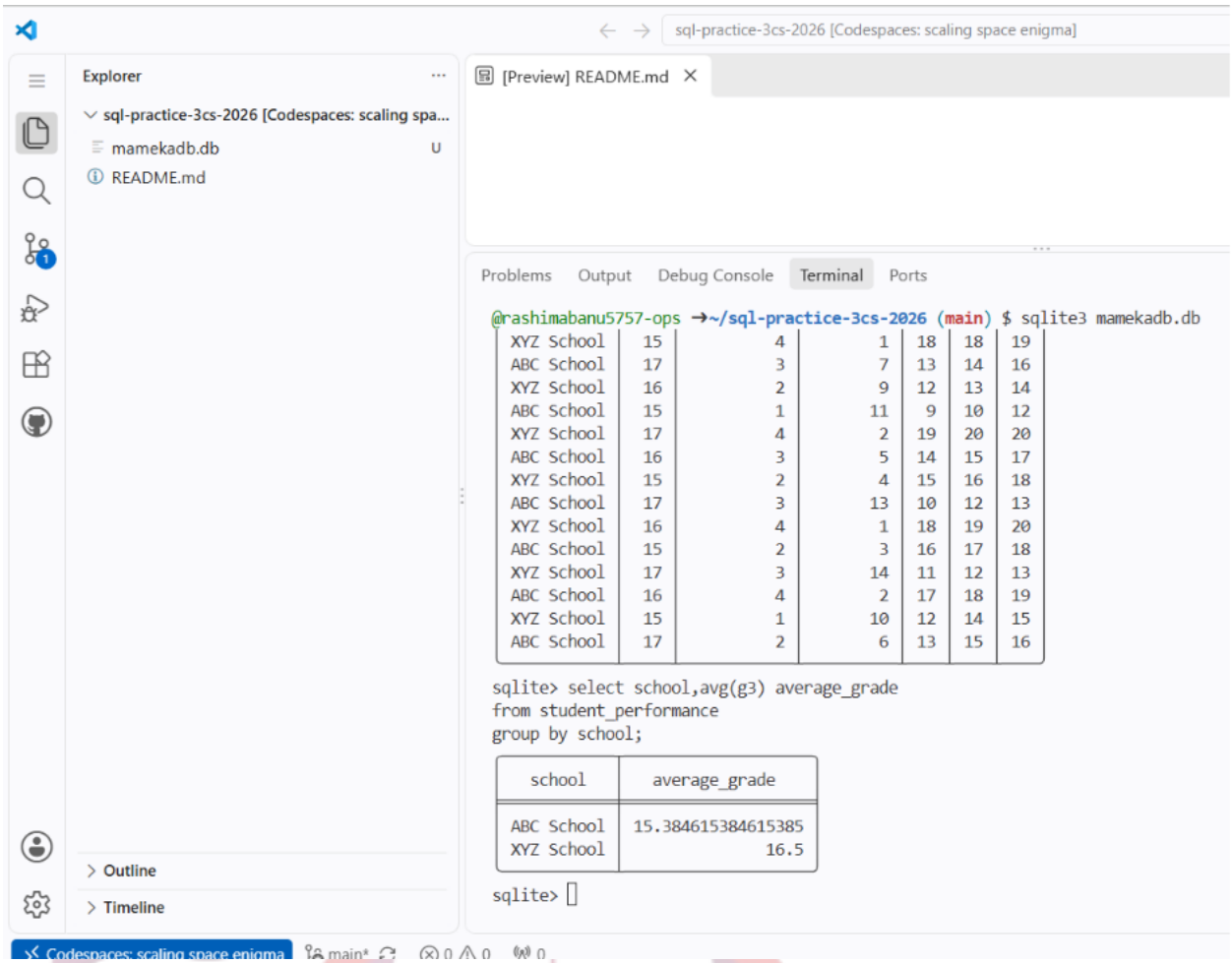
QUESTION: 2. Calculate the average final grade for each school

SQL> select school,avg(g3) average_grade

from student_performance

group by school;

OUTPUT:



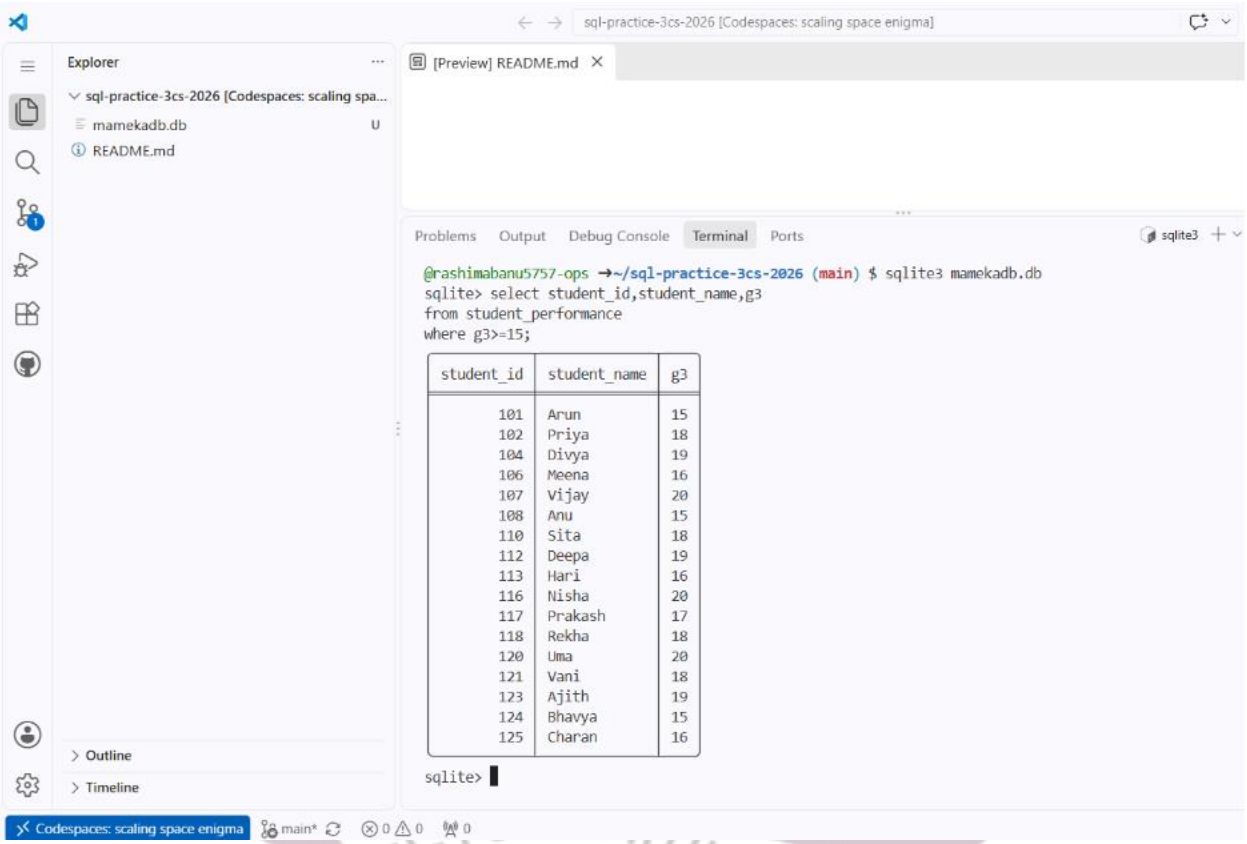
QUESTION: 3. Identify students who scored at least 15 in the final examination

SQL> select student_id,student_name,g3

from student_performance

where g3>=15;

OUTPUT:



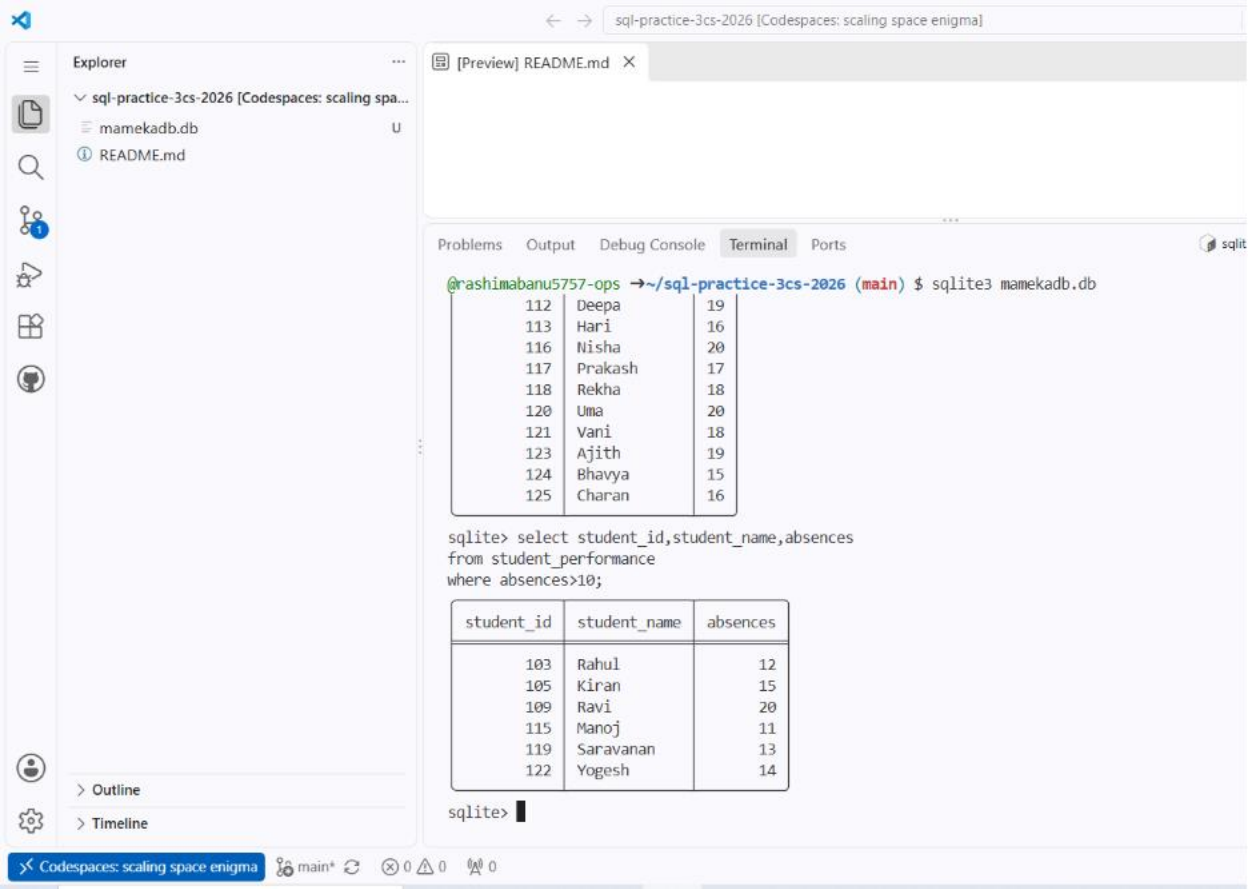
QUESTION: 4. Display students with more than 10 absences

SQL> select student_id,student_name,absences

from student_performance

where absences>10;

OUTPUT:

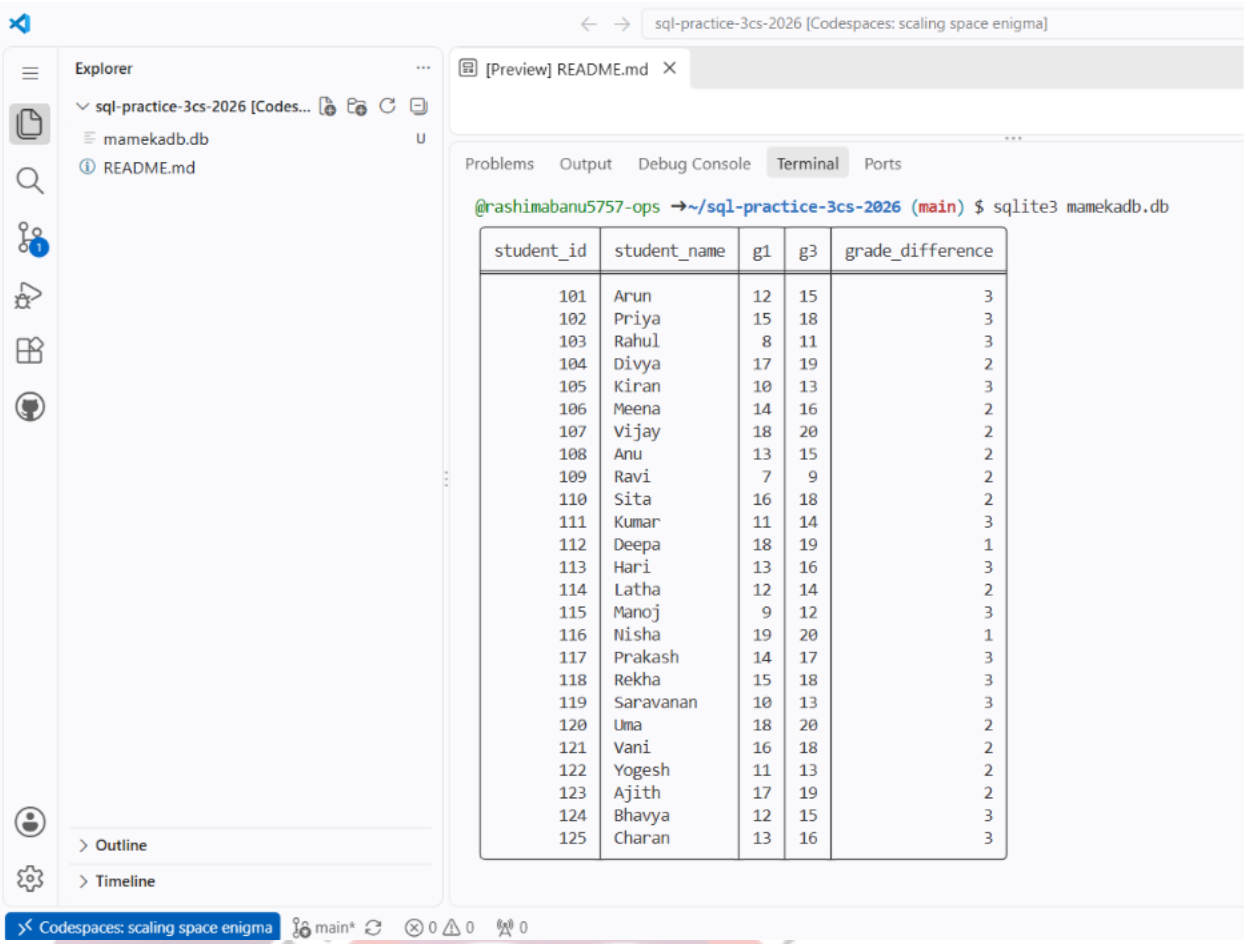


QUESTION: 5. Calculate the difference between first-period and final grades for each student

SQL> select student_id,student_name,g1,g3,(g3-g1) grade_difference

from student_performance;

OUTPUT:



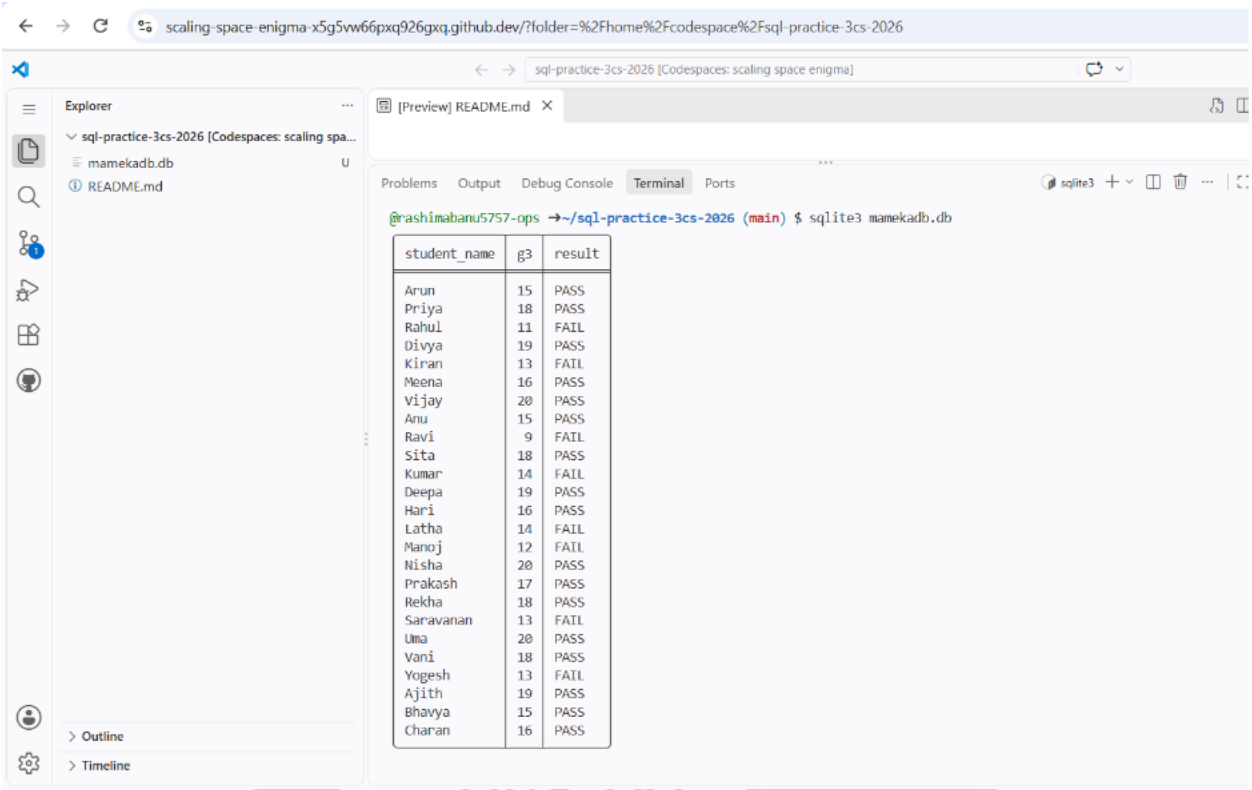
student_id	student_name	g1	g3	grade_difference
101	Arun	12	15	3
102	Priya	15	18	3
103	Rahul	8	11	3
104	Divya	17	19	2
105	Kiran	10	13	3
106	Meena	14	16	2
107	Vijay	18	20	2
108	Anu	13	15	2
109	Ravi	7	9	2
110	Sita	16	18	2
111	Kumar	11	14	3
112	Deepa	18	19	1
113	Hari	13	16	3
114	Latha	12	14	2
115	Manoj	9	12	3
116	Nisha	19	20	1
117	Prakash	14	17	3
118	Rekha	15	18	3
119	Saravanan	10	13	3
120	Uma	18	20	2
121	Vani	16	18	2
122	Yogesh	11	13	2
123	Ajith	17	19	2
124	Bhavya	12	15	3
125	Charan	13	16	3

PL/SQL TASKS

QUESTION: 1. Function to classify students based on final examination marks

SQL> select student_name,
g3,
case
when g3 >= 15 then 'PASS'
else 'FAIL'
end as result
from student_performance;

OUTPUT:



student_name	g3	result
Arun	15	PASS
Priya	18	PASS
Rahul	11	FAIL
Divya	19	PASS
Kiran	13	FAIL
Meena	16	PASS
Vijay	20	PASS
Anu	15	PASS
Ravi	9	FAIL
Sita	18	PASS
Kumar	14	FAIL
Deepa	19	PASS
Hari	16	PASS
Latha	14	FAIL
Manoj	12	FAIL
Nisha	20	PASS
Prakash	17	PASS
Rekha	18	PASS
Saravanan	13	FAIL
Uma	20	PASS
Vani	18	PASS
Yogesh	13	FAIL
Ajith	19	PASS
Bhavya	15	PASS
Charan	16	PASS

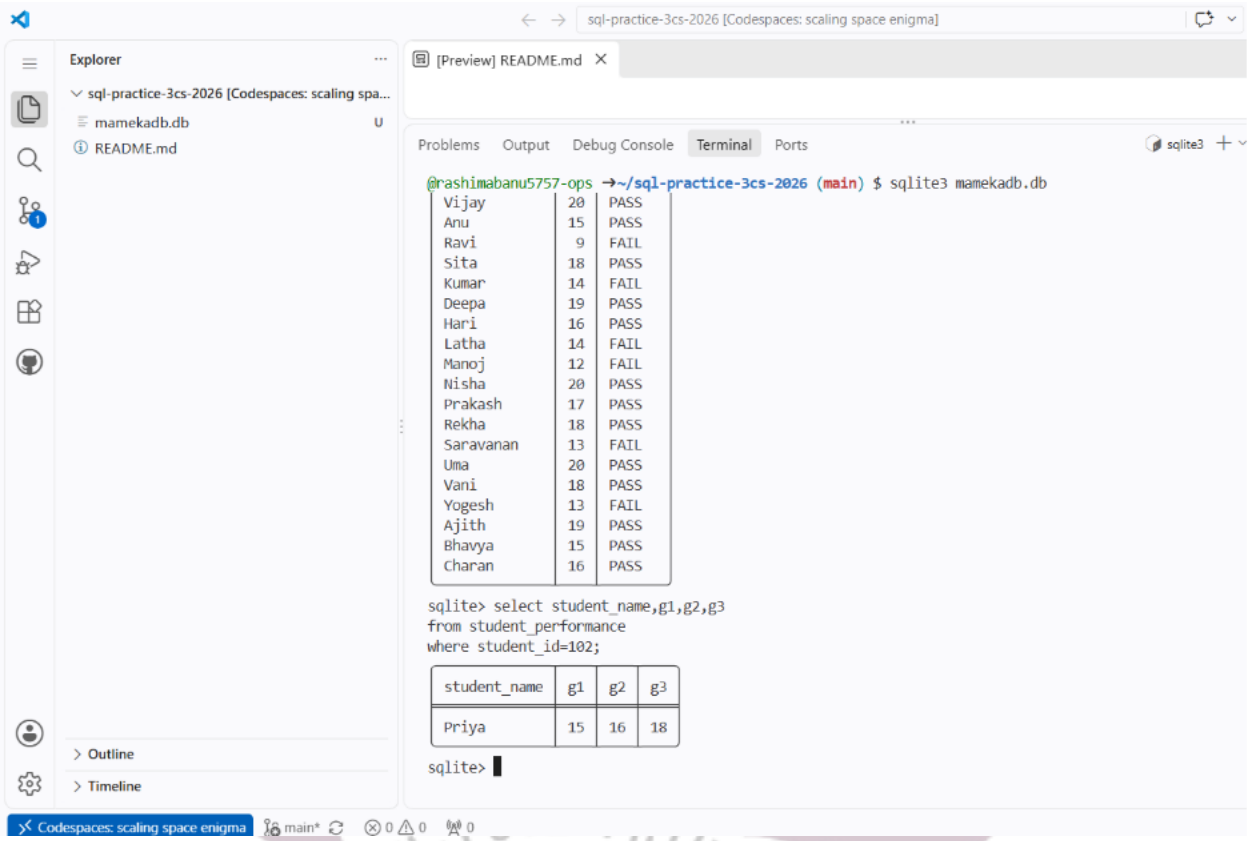
QUESTION: 2. Procedure to display academic performance using student ID

SQL> select student_name,g1,g2,g3

from student_performance

where student_id=102;

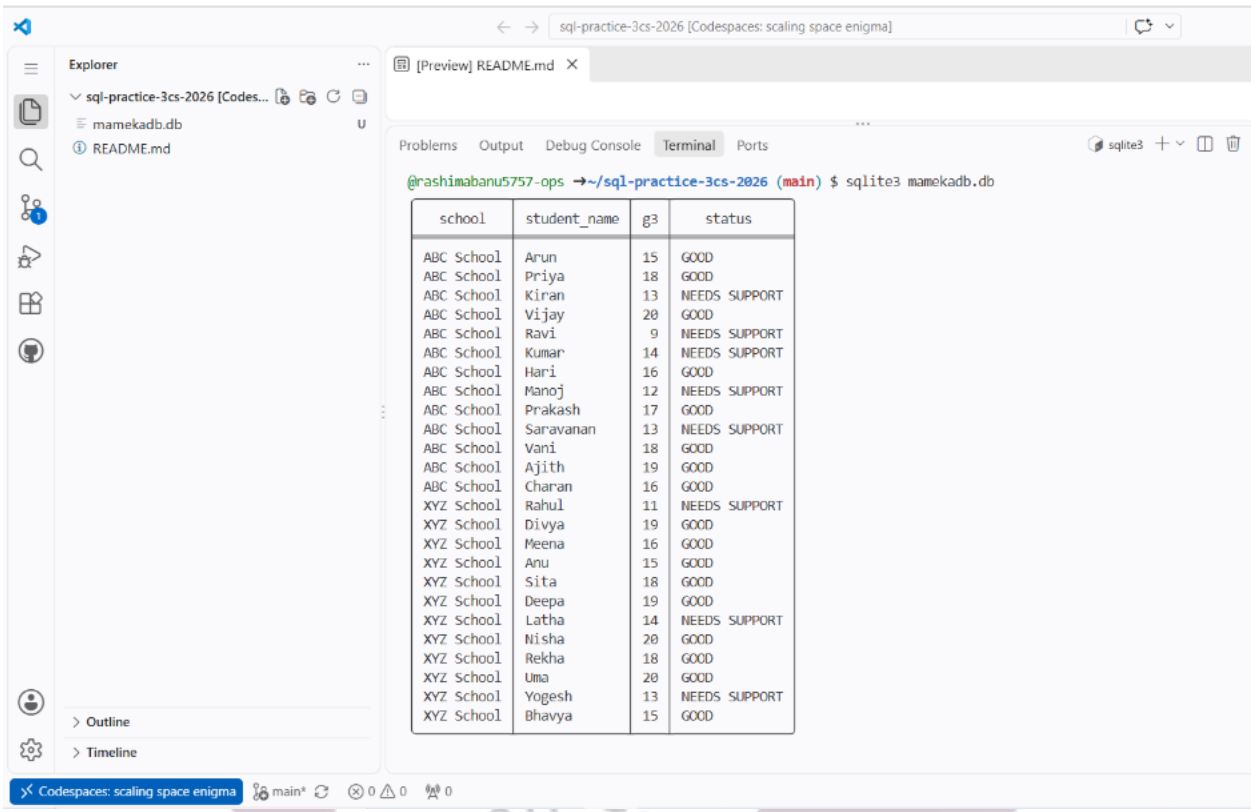
OUTPUT:



QUESTION: 3. Procedure to generate a school-wise performance report and identify students requiring academic support

SQL> select school,
student_name,
g3,
case
when g3 < 15 then 'NEEDS SUPPORT'
else 'GOOD'
end as status
from student_performance
order by school;

OUTPUT:



school	student_name	g3	status
ABC School	Arun	15	GOOD
ABC School	Priya	18	GOOD
ABC School	Kiran	13	NEEDS SUPPORT
ABC School	Vijay	20	GOOD
ABC School	Ravi	9	NEEDS SUPPORT
ABC School	Kumar	14	NEEDS SUPPORT
ABC School	Hari	16	GOOD
ABC School	Manoj	12	NEEDS SUPPORT
ABC School	Prakash	17	GOOD
ABC School	Saravanan	13	NEEDS SUPPORT
ABC School	Vani	18	GOOD
ABC School	Ajith	19	GOOD
ABC School	Charan	16	GOOD
XYZ School	Rahul	11	NEEDS SUPPORT
XYZ School	Divya	19	GOOD
XYZ School	Meena	16	GOOD
XYZ School	Anu	15	GOOD
XYZ School	Sita	18	GOOD
XYZ School	Deepa	19	GOOD
XYZ School	Latha	14	NEEDS SUPPORT
XYZ School	Nisha	20	GOOD
XYZ School	Rekha	18	GOOD
XYZ School	Uma	20	GOOD
XYZ School	Yogesh	13	NEEDS SUPPORT
XYZ School	Bhavya	15	GOOD

Exception Handling

Test Case 1 – Student Not Found

SQL> begin
student_report(999);
exception
when no_data_found then
dbms_output.put_line('Student ID not found');
when others then
dbms_output.put_line('Unexpected error occurred');
end;
/

OUTPUT:

status
STUDENT ID NOT FOUND

Test Case 2 – Invalid Final Grade

SQL> select student_name,
g3,
case
when g3 is null then 'ERROR'
when g3>=15 then 'PASS'
else 'FAIL'
end as result
from student_performance;

OUTPUT:

```
sqlite> select student_name,
       g3,
       case
           when g3 is null then 'ERROR'
           when g3>=15 then 'PASS'
           else 'FAIL'
       end as result
from student_performance
where student_id=126;
```

student_name	g3	result
Test Student	NULL	ERROR

```
sqlite>
```

Test Case 3 – School-wise Report with Academic Support

SQL> select case
when exists (select 1 from student_performance)
then 'School report generated successfully'
else 'Error while generating school report'
end as message;

OUTPUT:

```
sqlite> select case
when exists (select 1 from student_performance)
then 'School report generated successfully'
else 'Error while generating school report'
end as message;
```

message
School report generated successfully

```
sqlite>
```

Dataset Insights

- The dataset contains **25 student performance records**.
- The records contain student name, school, age, study time, absences, and examination grades.
- Final examination performance is represented by **G3**.
- Students scoring **15 or above** are considered to have good performance.
- Students scoring **below 15** are identified as requiring **academic support**.
- Invalid student IDs are handled using the `NO_DATA_FOUND` exception.
- Other unexpected errors are handled using the `WHEN OTHERS` exception.
- School-wise analysis helps identify students who require additional academic support.

Result

Thus, the **Student Performance Analysis System** was successfully implemented using **SQL and PL/SQL**. The required student records were stored and analyzed using SQL queries. PL/SQL functions and procedures were used to classify student performance, display academic details, and generate school-wise reports. **Exception handling was implemented successfully to handle invalid student IDs, missing data, and unexpected errors.**